Lab Assessment 2

Health Analytics Report

Insurance Medical Cost Prediction Using Linear Regression Model

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Table of Contents

List of Figures:

List of Tables:

1. Introduction

Accurate prediction of medical expenditures constitutes a critical component in both health insurance management and public health policy. Robust predictive models enable insurers to estimate risk more precisely, thereby informing equitable premium structures, while providing policymakers with evidence to design and implement targeted preventive interventions that mitigate future healthcare costs. This study centers on demographic, lifestyle, and geographic characteristics in conjunction with individual medical charges, providing a suitable basis for modelling healthcare expenditures and examining the key factors influencing insurance costs (Morid, et al., 2018).

Linear regression models are widely used in predictive tasks that involve multiple explanatory variables. Within the context of health insurance, their use is well established due to their simplicity and interpretability (Manning & Mullay, 2021). Nevertheless, healthcare cost data, such as medical charges, are typically right-skewed, which may introduce bias and undermine the core assumptions of ordinary least squares regression. To address this limitation, a logarithmic transformation of the cost variable is commonly employed to reduce skewness, achieve a more symmetric distribution, and improve linearity (Manning & Mullay, 2021).

In this study, the insurance dataset is analyzed from both exploratory and inferential perspectives to apply a linear regression model to predict medical charges. The analysis focuses on selecting features with the strongest associations to the dependent variable, followed by systematic evaluation of variable distributions, model performance, and predictive accuracy.

2. Methods

2.1 Dataset Description, Pre-processing, and Exploratory Data Analysis

The dataset comprises 1338 observations, with seven features that relate to the insurer: age, sex, body mass index (BMI), number of children, smoker status, region, and annual medical charges. **Table 1** shows the first five rows of the dataset.

Table 1*. Insurance cost dataset preview*

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16684.924
1	18	male	33.770	1	no	southeast	1725.552
2	28	male	33.000	3	no	southeast	4449.462
3	33	male	22.705	0	no	northwest	21984.471
4	32	male	28.880	0	no	northwest	3866.855

Medical charges exhibit significant positive skewness and a presence of large outliers, as shown in **Figure 2**(c). Therefore, a logarithmic transformation is applied to achieve a normal distribution of the dependent variable, as seen in **Figure 2**(h). This is done to preserve the linearity required for accurate linear regression model applications, as well as offer normally distributed residuals (Manning & Mullay, 2021).To achieve overall regression framework inclusion, one-hot encoding was applied to categorical variables such as sex, smoker, and region. The data had no missing values, satisfying best practices required for standard regression modelling (ul Hassan, et al., 2021).

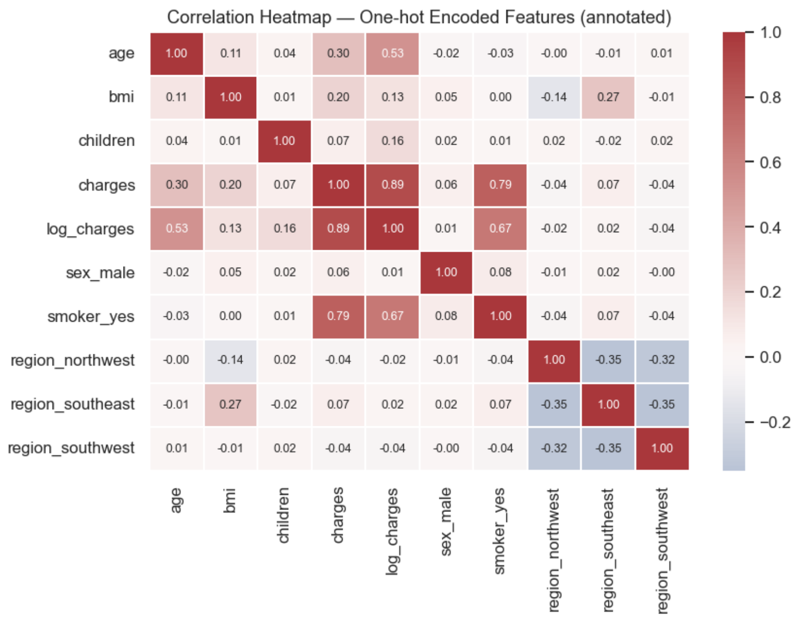


Figure 1*.Correlation Heatmap of features*

For the Correlation analysis, **Figure 1** shows the heatmap of the features, indicating that smoking status is the dominant predictor of medical charges, followed by age, while BMI contributes less moderately. Other data features, such as sex, number of children, and region of residence, display negligible correlation with charges. Therefore, for this model, age, BMI, and smoker status are selected as features to design the multiple linear regression model. These insights are consistent with prior empirical evidence in health economics, where lifestyle risk factors and age are primary cost drivers, whereas demographic factors such as gender and region tend to exert limited direct effects (ul Hassan, et al., 2021).

2.2 Linear Regression Model Specification and Evaluation

Using the sklearn library, a 75/25 random train-test split was used for model development and validation. The multiple linear regression model was based on numerical features age and bmi, as well as one-hot encoded categorical feature smoker. The other features, such as sex, children, and region, were excluded from the model training and evaluation because they showed insignificant correlation to medical charges. Performance metrics selected for this model included the coefficient of determination (a), mean absolute error (MSE), and root mean squared error (RMSE), applied on both original scale and logarithmic transformation (Manning & Mullay, 2021).

3. Results

3.1 Descriptive Results

Table 2 details descriptive statistics of input features. The average age was 39 years, with a range of 18-64 years. The BMI average was 30.7, indicating overweight. 20% of the population was found to be smokers. Medical charges ranged from 1,122 to 63,770 USD, confirming the right-skewed nature necessitating a logarithmic transformation.

Table 2.*Numeric Variables Descriptive Statistics*

	age	bmi	children	charges
COUNT	1338.0	1338.0	1338.0	1338.000
MEAN	39.2	30.7	1.1	13270.422
STD	14.0	6.1	1.2	12110.011
MIN	18.0	16.0	0.0	1121.874
25%	27.0	26.3	0.0	4740.287
50%	39.0	30.4	1.0	9382.033
75%	51.0	34.7	2.0	16639.913
MAX	64.0	53.1	5.0	63770.428

Figure 2(a)-(d) shows the histogram that indicates the distribution of sex, smokers, medical charges, and age, respectively. From **Figure 2(f)**, it is evident that people with smoking behaviour spend more on medical charges, while people who smoke less are charged less. The violin plot of sex in **Figure 2 (e)** shows that the distribution of male and female charges is similar for both. **Figure 2(g)** shows three bands of distribution, with the higher band likely representing smoking women, while the lower band strip represents smoking men.

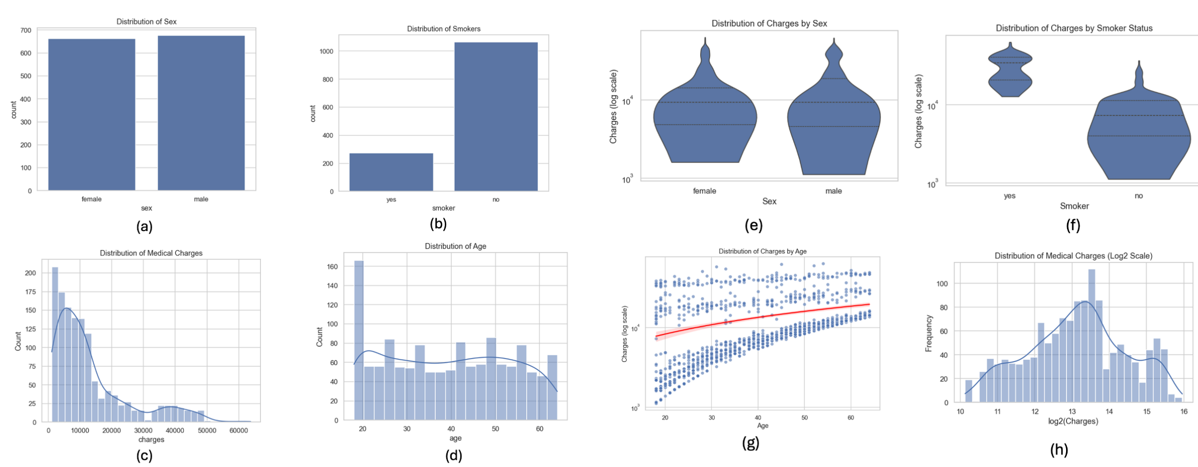
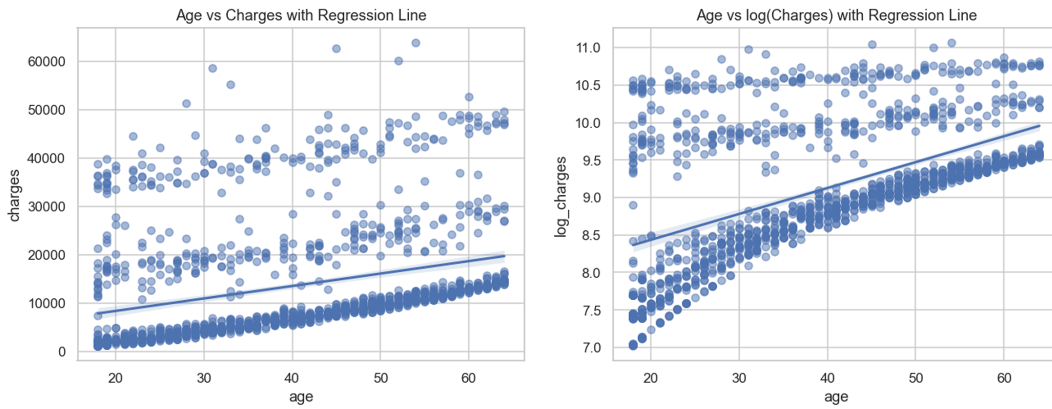


Figure 2 *(a-h) Distribution Plots of Sex, Age, Smoker Status, and Medical Charges*

Figure 3(a-b) shows graphs of a regression line before and after logarithmic transformation, which were plotted to explore the relationship between age and charges. The scatterplots revealed heteroscedasticity and non-constant variance in the raw age-charges relationship, which significantly improved after log transformation. This aligns with linear regression methodological framework guidance advocating for transformation for cost data modelling (Xu, et al., 2015).



(a) (b)

Figure 3 (a-b) *Age vs Medical Costs Regression line before and after Logarithmic Transformation.*

3.2 Correlation Analysis

The correlation heatmap (Figure 1) revealed that being a smoker had the strongest correlation to medical charges (r≈0.79). Age indicated a moderate correlation (r≈0.30), which increased to (r ≈ 0.53) when assessed against log-transformed charges.BMI showed a positive but weak correlation to charges (r≈= 0.2), while sex and region indicated a weak or negligible linear correlation to medical charges.

3.3 Model Performance

Table 3 shows the model performance metrics for the original and log-transformed medical charges scale. The standard scale metrics are in US Dollars.

Table 3. *Multiple Linear Regression model performance metrics*

	R^2	MAE	RMSE
Log-space metrics	0.748	0.302	0.466
Normal Scale metrics	0.362	5091	9813.74

4. Discussion

The results support the significant role of smoking in increasing medical insurance costs, reinforcing the epidemiological results quantifying the economic burden attributed to the use of tobacco. Age appears as a second driver, reflecting a substantial increase in morbidity among older adults. However, the moderate impact of BMI is in line with the literature supporting increased health care costs associated with obesity (Morid, et al., 2018).

Minimal correlation of age, sex, and region suggests that increased medical costs are more associated with behavioural and psychological factors, overshadowing demographic variables, as confirmed by previous findings (ul Hassan, et al., 2021).

Applying a log transformation on the medical costs effectively mitigates heteroscedasticity, corresponding with the methodological recommendations of (ul Hassan, et al., 2021) to achieve a more stable model, precise coefficients, and improved overall model accuracy and fit, especially in datasets with such extreme values in the data range. For transparency and healthcare policy implementation, linear regression interpretability offers an advantage (Manning & Mullay, 2021).

The linear regression model achieved an R-squared (R^2) score of 0.75, a Mean Absolute Error (MAE) of 5091, and a Root Mean Squared Error (RMSE) of 9813. While an accuracy score of 0.74 is described as satisfactory, and even "impressive" in one instance (Patra et al., 2024), with an R-squared score of 0.74, other models, such as Stochastic Gradient Boosting (SGB) and Random Forest Regressor, offer higher scores of 0.90 and 0.80, respectively.

5. Conclusion

In Conclusion, a multiple linear regression model was applied to a log-transformed insurance cost dataset, highlighting smoker status and age of insurer as key drivers of medical insurance charges. BMI indicated a moderate impact, whereas other variables, such as number of children, sex, and region, showed a negligible contribution. Implementation of the log scale significantly improved the skewness in the original dataset, allowing for enhanced linear regression model performance ($R^2 = 0.74$). Even though key limitations such as deviations from normal distribution of charges, and a low accuracy as compared to other machine learning models, this study highlights the importance of linear regression as a prediction model and tool that can be utilised in health analytics, applicable for preliminary analysis and informing policymakers. It would be beneficial for future research to address issues such as outliers and skewed data, as well as addressing model assumptions to account for non-linear relationships between variables and cost.

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